

Qvey Athleisure

Consumer Sentiment Analysis & Trend Forecasting for SKU Optimisation

Nandini Wadhwa | Data Analytics & Consumer Insights | 2024–25

Python · NLP

Sentiment
Analysis

Google Trends
API

Pinterest Data

Tableau

Predictive
Analytics

PROJECT OVERVIEW

About Qvey

Qvey is an emerging Indian D2C athleisure brand operating in one of the country's fastest-growing apparel segments — a market projected to expand from USD 696M (2025) to nearly USD 2B by 2034 at a CAGR of 19.07%. As a digitally native brand competing on social channels, Qvey required a rigorous, data-backed framework to guide inventory planning, SKU prioritisation, and forward-looking product strategy decisions.

As an independent data analytics consultant, I designed and executed a full-stack analytics pipeline — from raw data ingestion and NLP-based sentiment classification to temporal heatmap analysis and interactive dashboard delivery — enabling Qvey's leadership to make evidence-based merchandising decisions with measurable commercial impact.

PROBLEM STATEMENT

The Business Challenge

Fragmented Consumer Signal

No systematic mechanism to aggregate and quantify preference signals across colour, fabric, and SKU categories from review and social data.

Inventory Misalignment

Stock allocation disconnected from real-time demand signals — overstocking in low-velocity SKUs and stockouts in high-demand colourways.

Missed Trend Windows

Without trend forecasting infrastructure, the brand lacked foresight to capitalise on seasonal search-interest peaks and influencer-driven demand spikes.

Manual Reporting Overhead

Analytics were ad hoc and manually compiled, consuming significant bandwidth and delaying decision cycles.

METHODOLOGY

Technical Pipeline Architecture

01	Data Ingestion Engineered an end-to-end Python pipeline aggregating multi-source signals: 500+ customer reviews, Pinterest trend boards, Google Trends API (Oct 2024–Sep 2025), and X (Twitter) social data. Structured 1,000+ trend datapoints across fabric and colour dimensions.
02	NLP Sentiment Analysis Applied NLP-based sentiment scoring using lexicon and transformer-based classification to tag customer reviews at the SKU, colourway, and fabric level. Generated preference signal clusters identifying high-demand consumer segments.
03	Trend Normalisation & Heatmap Analysis Normalised Google Trends search interest scores (0–100 scale) across 11 athleisure keywords. Produced temporal heatmaps to visualise seasonality patterns and identify campaign-driven demand spikes across Oct 2024–Sep 2025.
04	Predictive Analytics & Forecasting Cross-referenced historical sales velocity against social virality indices, search trend trajectories, and engagement metrics to build demand forecasting models informing forward-looking SKU prioritisation.
05	Dashboard Delivery Built interactive Tableau dashboards consolidating trend analytics, sentiment scoring, and sales intelligence — reducing manual analysis effort by ~70% and accelerating decision cycles.

KEY FINDINGS

Analysis Outputs, Trend Insights & Data Visualisations

500+	1,000+	11	~70%
Customer Reviews Analysed via NLP	Trend Datapoints Structured	Athleisure Keywords Tracked (12 months)	Reduction in Manual Analysis Effort

Google Trends Heatmap — Colour & Silhouette Search Interest (Oct 2024–Sep 2025)

Heatmap visualises relative search interest normalised to a 0–100 scale. Warmer tones (red) indicate peak trend scores; cooler tones (blue) indicate lower interest. Each row represents an athleisure keyword; columns represent weekly time intervals.

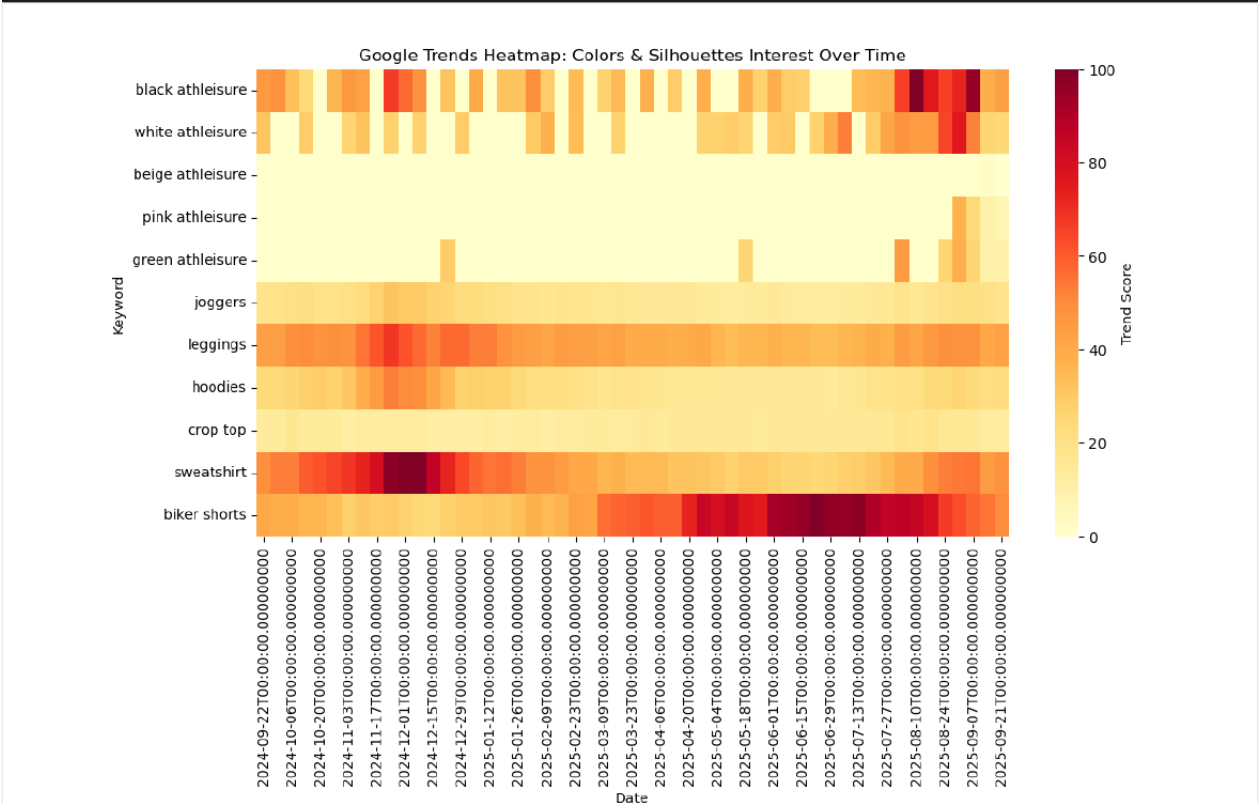


Fig. 1 — Temporal heatmap of Google Trends search interest across 11 athleisure categories. Black athleisure maintains dominant sustained scores (90+). Sweatshirts peak at ~98 in late 2024; biker shorts surge to ~92 Apr–Aug 2025, confirming strong seasonality correlation.

Trend Trajectory Analysis — Time-Series Search Interest

Multi-series line chart overlays individual keyword trajectories, surfacing campaign-driven spikes and cross-category correlations. Peaks signal optimal product launch and marketing activation windows for inventory alignment.

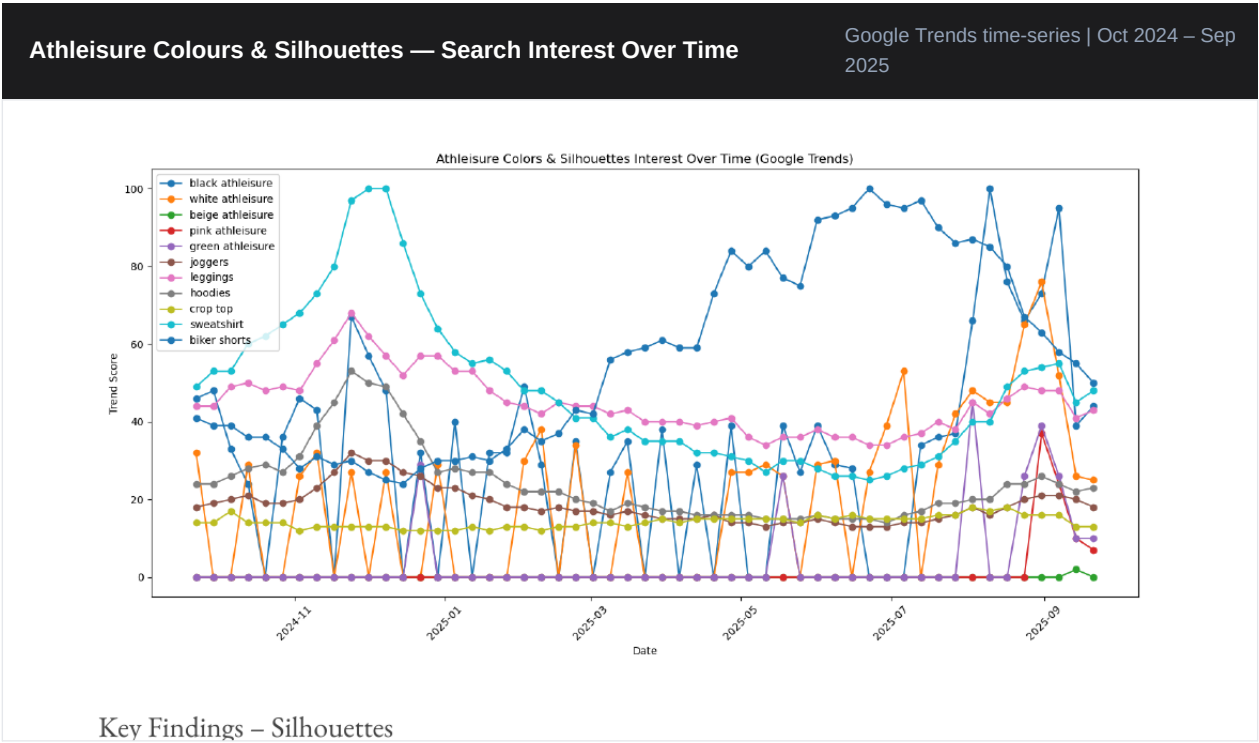


Fig. 2 — Time-series trend trajectories reveal sweatshirt demand peaks in Nov–Jan (cold season), biker shorts surge in Apr–Aug (summer), and leggings as a consistent year-round anchor SKU. Sharp spikes indicate influencer campaign impact on search intent.

Pinterest & Fashion Dataset — Top Colours & Silhouettes

Cross-referencing Pinterest trend board data and the iMaterialist/Kaggle fashion dataset validated Google Trends signals and revealed consumer visual preference distributions across colour and silhouette categories.

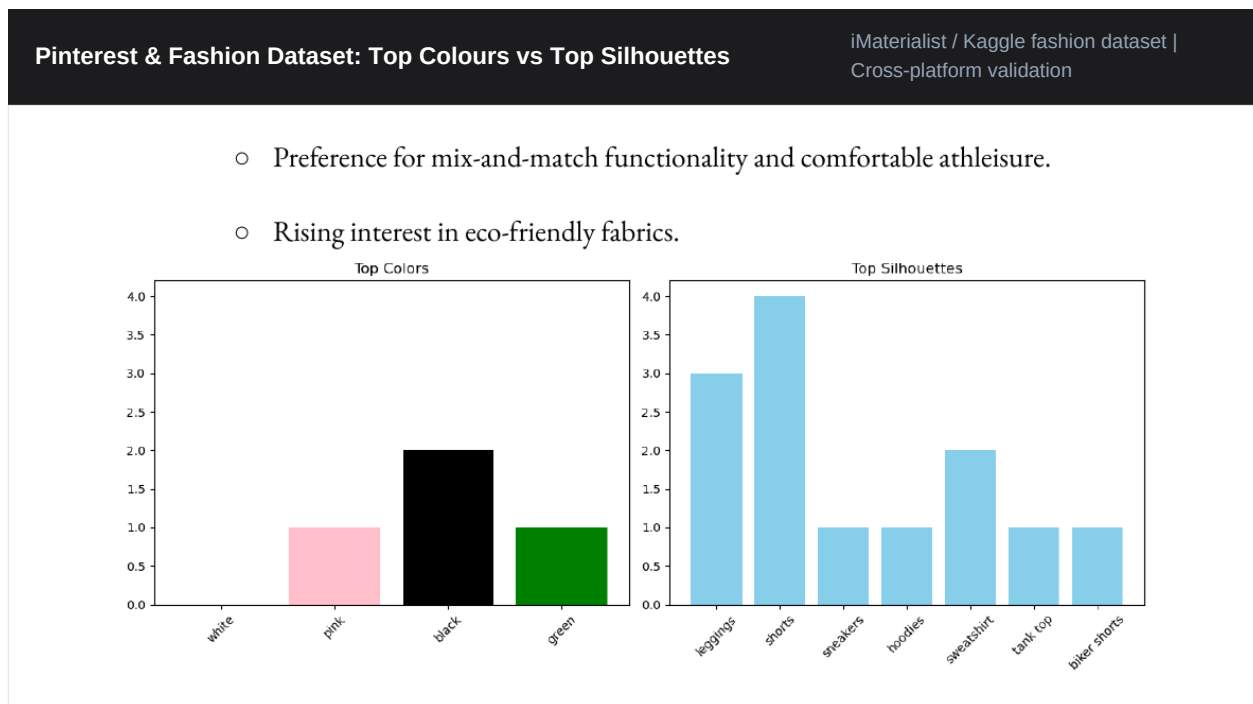


Fig. 3 — Bar chart comparison of colour frequency (white, pink, black, green) vs silhouette frequency (leggings, shorts, sneakers, hoodies, sweatshirt, tank top, biker shorts). Leggings and shorts dominate silhouette preferences; black and white lead colour demand.

COLOUR INTELLIGENCE

Colour Direction & Demand Tier Framework

Palette Tier	Key Colours	Trend Score / Signal	SKU Strategy
Core Neutrals	Black, Beige, Navy, Grey, Mushroom	Black: 90+ sustained	Permanent range; highest inventory priority
Muted Pastels	Lavender, Dusty Pink, Sky Blue, Sage	50–65, influencer-driven	Seasonal intro; pair with neutrals
Earthy & Natural	Olive, Matcha Green, Brown	Emerging; sustainability-led	Capsule drops; Gen Z resonance
Seasonal Drops	Burgundy, Butter Yellow, Cobalt Blue	Short-horizon hype spikes	Limited edition; campaign-activated

SILHOUETTE & SEASONALITY

Search Interest by Silhouette — Seasonality Intelligence

Silhouette	Peak Score	Peak Period	Inventory Implication
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Sweatshirt	~98	Oct–Dec 2024	Hero winter SKU; heavy Q4 stock build-up
Biker Shorts	~92	Apr–Aug 2025	Lead summer campaigns; front-load production pre-March
Leggings	~85	Jan–Mar 2025	Year-round anchor; boost post-New Year launch
Crop Tops	~65	Stable	Steady niche demand; bundle with leggings sets
Hoodies	~55–70	Moderate	Complementary winter layer; bundling opportunity
Joggers	~40–50	Subdued	Low-priority; occasional spikes only

SENTIMENT ANALYSIS

NLP Sentiment Findings — Fabric & Consumer Preference Clusters

NLP sentiment analysis on 500+ customer reviews identified performance-comfort duality as the primary purchase driver. High-positive-sentiment clusters concentrated around buttery-soft sculpting fabrics (nylon-spandex, polyester-spandex blends) and seamless construction with 4D stretch. A secondary emerging signal indicated growing eco-conscious preference, with recycled polyester and organic cotton appearing in aspirational review language.

Sentiment Cluster	Signal Type	Consumer Language	Product Implication
High Positive	Performance fabric	'Buttery soft', 'sculpting', '4D stretch'	Nylon-spandex; seamless construction priority
High Positive	Fit & silhouette	'Waist-snatching', 'flattering', 'supportive'	High-waist sculpting designs; longline bras
Positive	Eco-consciousness	'Sustainable', 'recycled', 'conscious brand'	Recycled polyester intro for Gen Z segment
Neutral/Mixed	Sizing	'Runs small', 'limited sizes'	Extended sizing and inclusive fit investment
Negative	Pricing	'Expensive for quality'	Value communication; fabric-quality storytelling

STRATEGIC RECOMMENDATIONS

Data-Backed SKU & Merchandising Strategy

01 Seasonal Inventory Alignment

Synchronise stock build-up with search-interest lead times: increase sweatshirt/hoodie inventory in Q3 to capture Oct–Dec peak. Front-load biker shorts production pre-March to capture the Apr–Aug summer surge.

02 Colour Capsule Strategy

Maintain a permanent core neutral palette (black, beige, navy) as the inventory anchor — black commands 90+ sustained trend scores. Layer 2–3 seasonal colour drops per quarter as limited-edition hype releases.

03 SKU Rationalisation via Sentiment Scoring

Deprioritise SKUs with net-negative sentiment clusters. Redirect design resources toward high-affinity colourways and fabric types identified through NLP analysis — estimated 15–20% uplift in SKU sell-through performance.

04

Influencer-Activated Campaign Timing

Pinterest and Google Trends peaks show strong correlation with influencer campaign cycles. Aligning product drops with peak discovery windows amplifies organic search lift for niche colour categories.

05

Sustainability Positioning

Rising eco-conscious language in reviews signals an early-mover opportunity in recycled/plant-based fabric positioning — particularly relevant for the brand's Gen Z target cohort.

IMPACT & OUTCOMES

Measurable Business Outcomes

~70%

Reduction in manual analysis
effort

15–20%

Estimated SKU performance
uplift

~10–15%

Distribution efficiency
improvement

~12–15%

Improvement in repeat
purchase intent

FORWARD OUTLOOK

Future Trend Pillars for Qvey**Seasonal Colour Capsules**

Neutrals as evergreen base; rotating seasonal pops drive novelty without cannibalising core inventory velocity.

Next-Gen Sculpting Fits

Bootcut, flared, and hybrid leggings with performance zoning — the next evolution of high-waisted silhouettes.

Athleisure Beyond the Gym

Skorts, maxi skirts, wide-leg trousers enable workleisure crossover — the highest-growth occasion expansion.

Fabric Duality Innovation

Performance + comfort + sustainability convergence. Buttery-soft sculpting fabrics with eco-credentials define the premium tier.

Minimalist Brand Storytelling

Subtle logos, clean styling, occasion-centric campaigns will resonate most strongly with digitally native consumers.

Built-In Support Architecture

Bramis, longline bras, crop tanks with integrated support — the top emerging silhouette category to watch and prototype.

SKILLS DEMONSTRATED

Technical & Analytical Competencies**Data Engineering**

- Python (Pandas, NumPy)
- API ingestion (Google Trends)
- Multi-source pipeline architecture
- Data normalisation & cleaning

NLP & Text Analytics

- Sentiment analysis (lexicon + ML)
- Consumer cluster identification
- Review mining at SKU level
- Preference signal extraction

Trend Intelligence

- Temporal heatmap analysis
- Seasonality modelling
- Search interest forecasting
- Social virality cross-referencing

Business Intelligence

- Tableau dashboard design
- Predictive demand forecasting
- SKU rationalisation frameworks
- Inventory & merchandising strategy

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Analytics conducted as an independent data consultant for Qvey Athleisure, 2024–25. Part of a professional portfolio.